Amplifier calculations

U2 is an MCP6N16-100 instrumentation amplifier. It takes the bridge signal and sets the output with two resistors. The figures below use gain 151.4.

$$V_{OUT} = V_{REF} + G \times V_{IN} \qquad G = 1 + \frac{R2}{R1}$$

Choosing the gain

The biggest input is 3000 με × 4.29 μV per με = 12.87 mV.

Limit	Calculation	Gain
Lowest stable gain (datasheet)	Fixed by the 100 version of the part	100
Highest gain before the output clips	(3.25 V – 0.825 V) / 12.87 mV	188
Our choice, R2 = 100 kΩ, R1 = 665 Ω	1 + 100k / 665	151.4

We sit between the two limits, so the output has room left for bridge imbalance.

Output

Input	Calculation	Output
Per με	4.29 μV × 151.4	0.649 mV
3000 με compression	0.825 V + 12.87 mV × 151.4	2.77 V
300 με extension	0.825 V – 1.29 mV × 151.4	0.63 V

Checks against the datasheet

Check	Calculation	Result	Limit
Input size	12.87 mV	12.9 mV	27 mV
Bandwidth	350 kHz × 100 / 151.4	231 kHz	Above the 480 Hz filter
R2 stability	0.25 × 151.4² / (2π × 35 MHz × 16 pF)	1.6 MΩ limit	R2 is 100 kΩ
Noise	45 nV/√Hz × √(1.57 × 480 Hz)	1.24 μV rms, 0.29 με	Below one ADC count
Offset	17 μV / 4.29 μV per με	4 με maximum	Removed by calibration
Gain error	0.05% + 0.02% + 0.15%	0.22% maximum	Removed by calibration

The 16 pF is 14 pF of amplifier input capacitance plus 2 pF of board stray, which is our estimate. The 4.29 μV per με assumes 3.3 V excitation, gauge factor 2, and Poisson ratio 0.3.